

Ex 29:

① $Z = \max\{X, Y\}, Y = \min\{X, Y\}.$

② X, Y are independent.

$$F_Z(z) = F_X(z)F_Y(z)$$

$$f_Z(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z)$$

$$(1 - F_W(w)) = (1 - F_X(w))(1 - F_Y(w))$$

$$f_W(w) = f_X(w)(1 - F_Y(w)) + f_Y(w)(1 - F_X(w)).$$

③

$X, Y \sim \exp(\lambda).$ X, Y are independent.

$$f_X(x) = \lambda e^{-\lambda x} \\ = f_Y(x).$$

$$W = \min\{X, Y\} \Rightarrow f_W(w) = \lambda e^{-\lambda w} \lambda e^{-\lambda w} \quad F_X(x) = 1 - e^{-\lambda x}.$$

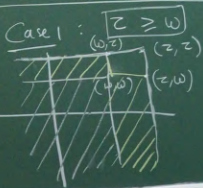
$$W \sim \exp(2\lambda)$$

$$+ \lambda e^{-\lambda w} e^{-\lambda w} = 2\lambda e^{-2\lambda w}.$$

⑥ X, Y have joint pdf $f_{X,Y}(x,y)$
and joint CDF $F_{X,Y}(x,y)$

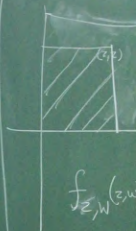
Find the joint pdf of Z, W $f_{Z,W}(z,w)$

$$\begin{aligned} F_{Z,W}(z,w) &= P(Z \leq z, W \leq w) \\ &= P(\max(X,Y) \leq z, \min(X,Y) \leq w) \\ &\Rightarrow \text{both } X, Y \leq z, \text{ at least one of } X, Y \leq w \end{aligned}$$

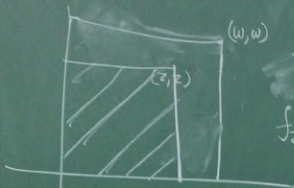


$$\begin{aligned} F_{Z,W}(z,w) &= F_{X,Y}(w,z) + F_{X,Y}(z,w) - F_{X,Y}(w,w) \\ f_{Z,W}(z,w) &= \frac{\partial^2 F_{Z,W}(z,w)}{\partial z \partial w} = f_{X,Y}(w,z) + f_{X,Y}(z,w) \end{aligned}$$

Case 2: $z < w$



Case 2: $z < w$



$$F_{Z,W}(z,w) = F_{X,Y}(z,z)$$

$$f_{Z,W}(z,w) = 0 \quad z < w$$

$$f_{Z,W}(z,w) = \begin{cases} f_{X,Y}(z,w) + f_{X,Y}(w,z) & z \geq w \\ 0 & z < w \end{cases}$$

$$+ F_{X,Y}(z,w) - F_{X,Y}(w,w)$$

$$f_{X,Y}(w,z) + f_{X,Y}(z,w)$$

Suppose X, Y are independent & identical

$$f_{Z,W}(z,w) = \begin{cases} 2f_X(z)f_X(w) & z \geq w \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = f_Y(y) = f(r)$$

$$f_{X,Y}(z,w)$$

Ordered Statistics

$$X_1, X_2, \dots, X_n$$

$$Y_1 = \text{Smallest of } X_1, X_2, \dots, X_n \\ = \min(X_1, X_2, \dots, X_n)$$

$$Y_2 = \text{second smallest of } X_1, \dots, X_n$$

$$Y_i = i\text{-th term that appears in} \\ \text{ascending order of } X_1, \dots, X_n$$

$$Y_n = \text{largest of } X_1, \dots, X_n \\ = \max(X_1, \dots, X_n)$$

$$f_{Y_1, \dots, Y_n}(y_1, y_2, \dots, y_n) = ?$$

$$f_X(x) = f_Y(x) \\ = f(x)$$

$\Uparrow$

not identical

$$Z \geq W$$

otherwise

Ordered Statistics

$$X_1, X_2, \dots, X_n$$

$$Y_1 = \text{Smallest of } X_1, X_2, \dots, X_n \\ = \min(X_1, X_2, \dots, X_n)$$

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$$Y_i = i\text{-th term that appears in} \\ \text{ascending order of } X_1, \dots, X_n$$

$$Y_n = \text{largest of } X_1, \dots, X_n \\ = \max(X_1, \dots, X_n)$$

Assuming $X_1, \dots, X_n$
are independent & identical

$$f_{Y_1, \dots, Y_n}(y_1, y_2, \dots, y_n) = \begin{cases} n! \prod_{i=1}^n f(y_i) & y_1 < y_2 < \dots < y_n \\ 0 & \text{otherwise} \end{cases}$$

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$$P\left(y_1 - \frac{\epsilon}{2} \leq Y_1 \leq y_1 + \frac{\epsilon}{2}, y_2 - \frac{\epsilon}{2} \leq Y_2 \leq y_2 + \frac{\epsilon}{2}, \dots, y_n - \frac{\epsilon}{2} \leq Y_n \leq y_n + \frac{\epsilon}{2}\right)$$

$$= \sum_{\substack{(i_1, \dots, i_n) \\ \text{are a permutation} \\ \text{of } (1, 2, \dots, n)}} P\left(y_1 - \frac{\epsilon}{2} \leq X_{i_1} \leq y_1 + \frac{\epsilon}{2}, \dots, y_n - \frac{\epsilon}{2} \leq X_{i_n} \leq y_n + \frac{\epsilon}{2}\right)$$

Assume $n=2$

$$y_1 - \frac{\epsilon}{2} \leq Y_1 \leq y_1 + \frac{\epsilon}{2}, y_2 - \frac{\epsilon}{2} \leq Y_2 \leq y_2 + \frac{\epsilon}{2}$$

$$y_1 < y_2 < \dots < y_n$$

$$\epsilon \leq \min_{i \in \{2, \dots, n\}} (y_i - y_{i-1})$$

$$\iff y_1 - \frac{\epsilon}{2} \leq X_1 \leq y_1 + \frac{\epsilon}{2}, y_2 - \frac{\epsilon}{2} \leq X_2 \leq y_2 + \frac{\epsilon}{2}$$

or

$$y_2 - \frac{\epsilon}{2} \leq X_1 \leq y_2 + \frac{\epsilon}{2}, y_1 - \frac{\epsilon}{2} \leq X_2 \leq y_1 + \frac{\epsilon}{2}$$

$$y_1 + \frac{\epsilon}{2} \leq y_2 - \frac{\epsilon}{2}$$

$$\epsilon \leq y_2 - y_1$$

RHS.

$$= \sum_{(i_1, \dots, i_n) \in S_n}$$

representation
of permutations
on $(1, 2, \dots, n)$

$$|S_n| = n!$$

$$\text{LHS} = f_{Y_1, \dots, Y_n}(\bar{y}_1, \dots, \bar{y}_n) \varepsilon^n$$

$$f_{X_1, \dots, X_n}(y_1, y_2, \dots, y_n) \varepsilon^n$$

$$\begin{aligned} & P(y_1 \leq Y_1 \leq y_1 + \varepsilon, \dots, y_n \leq Y_n \leq y_n + \varepsilon) \\ &= \int_{y_1 - \varepsilon}^{y_1 + \varepsilon} \dots \int_{y_n - \varepsilon}^{y_n + \varepsilon} f_{Y_1, \dots, Y_n}(u_1, \dots, u_n) du_1 \dots du_n \\ &\approx f_{Y_1, \dots, Y_n}(\bar{y}_1, \dots, \bar{y}_n) \int_{y_1 - \varepsilon}^{y_1 + \varepsilon} \dots \int_{y_n - \varepsilon}^{y_n + \varepsilon} du_1 \dots du_n = \varepsilon^n \end{aligned}$$

$$\Rightarrow f_{X_1, \dots, X_n}(y_1, \dots, y_n) = \sum_{\sigma \in S_n} f_{Y_1, \dots, Y_n}(y_{\sigma(1)}, y_{\sigma(2)}, \dots, y_{\sigma(n)})$$

If $X_1, \dots, X_n$ are identical & permutation independent

$$\begin{aligned} &= \sum_{\sigma \in S_n} \prod_{i=1}^n f_{X_i}(y_{\sigma(i)}) \stackrel{\text{identical } X_i\text{'s}}{=} \sum_{\sigma \in S_n} \left(\prod_{i=1}^n f(y_{\sigma(i)}) \right) = |S_n| \prod_{i=1}^n f(y_i) = n! \prod_{i=1}^n f(y_i) \end{aligned}$$

$n=2$
 X_1, X_2 if X_1, X_2 are independent

$$P(y_1 \leq Y_1 \leq y_1 + \varepsilon, y_2 \leq Y_2 \leq y_2 + \varepsilon)$$

$$\Downarrow$$

$$X_1, X_2$$

$$P(X_1 = X_2) = 0$$

$$P(X_1 = X_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

$$\prod_{i=1}^n f(y_i) \left[\lim_{\varepsilon \rightarrow 0} P(X_1 \leq X_1 + \varepsilon, X_2 \leq X_2 + \varepsilon) \right]$$

RHS

$$= \sum_{(i_1, \dots, i_n) \in S_n}$$

representation
of permutations
on $(1, 2, \dots, n)$

$$|S_n| = n!$$

$$\text{LHS} = f_{X_{i_1}, \dots, X_{i_n}}(y_1, \dots, y_n) \varepsilon^n$$

$$f_{X_{i_1}, \dots, X_{i_n}}(y_1, y_2, \dots, y_n) \varepsilon^n$$

$$\begin{aligned} & P(y_{i_1} \leq Y_1 \leq y_1 + \varepsilon, \dots, y_{i_n} \leq Y_n \leq y_n + \varepsilon) \\ &= \int_{y_{i_1} - \varepsilon}^{y_1 + \varepsilon} \dots \int_{y_{i_n} - \varepsilon}^{y_n + \varepsilon} f_{X_{i_1}, \dots, X_{i_n}}(u_1, \dots, u_n) du_1 \dots du_n \\ &\approx f_{X_{i_1}, \dots, X_{i_n}}(y_1, \dots, y_n) \int_{y_{i_1} - \varepsilon}^{y_1 + \varepsilon} \dots \int_{y_{i_n} - \varepsilon}^{y_n + \varepsilon} du_1 \dots du_n \quad \varepsilon^n \end{aligned}$$

$$\Rightarrow f_{X_{i_1}, \dots, X_{i_n}}(y_1, \dots, y_n) = \sum_{\sigma \in S_n} f_{X_{i_1}, \dots, X_{i_n}}(y_{\sigma(1)}, y_{\sigma(2)}, \dots, y_{\sigma(n)})$$

If $X_1, \dots, X_n$ are identical & permutation independent

$$\begin{aligned} &= \sum_{\sigma \in S_n} \prod_{i=1}^n f_{X_{i_1}}(y_{\sigma(i)}) \stackrel{\text{identical } X_i\text{'s}}{=} \sum_{\sigma \in S_n} \left(\prod_{i=1}^n f(y_{\sigma(i)}) \right) = |S_n| \prod_{i=1}^n f(y_i) = n! \prod_{i=1}^n f(y_i) \end{aligned}$$

1.

1.2
 X_1, X_2 if X_1, X_2 are independent

$$P(y_1 \leq Y_1 \leq y_1 + \varepsilon, y_2 \leq Y_2 \leq y_2 + \varepsilon)$$

$$\Downarrow$$

$$X_1, X_2$$

$$P(X_1 = x) = 0$$

$$P(X_1 = x_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1, X_2}(u_1, u_2) du_1 du_2$$

$$\prod_{i=1}^n f(y_i) \stackrel{= \lim_{\varepsilon \rightarrow 0} P(X_1 \leq x_1 \leq x_1 + \varepsilon)}{=}$$

Let

X is discrete

$$M_X(s) = E[e^{sx}]$$

$$= \sum_{n \in \mathcal{X}_0} e^{sn} p(n) \quad \left. \vphantom{\sum_{n \in \mathcal{X}_0}} \right\} \text{ This is similar to } Z\text{-transform where } Z = e^s.$$

X is continuous

$$M_X(s) = \int_{-\infty}^{\infty} e^{sx} f_X(x) dx$$

Similar to Laplace transform.

PGF

$\Updownarrow$ Discrete R.V.

PMF

$$G_X(z) = E[z^X] \quad |z| \leq 1$$

Examples

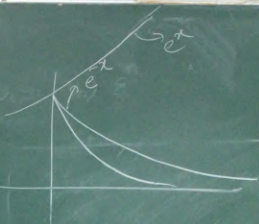
(a) $X \sim \text{exp}(\lambda)$

$$M_X(s) = \int_0^{\infty} e^{sx} \lambda e^{-\lambda x} dx$$

$$= \int_0^{\infty} \lambda e^{-(\lambda-s)x} dx$$

$$= \begin{cases} \frac{-1}{\lambda-s} e^{-(\lambda-s)x} \Big|_0^{\infty} & \lambda-s > 0 \\ \text{not defined} & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{\lambda}{\lambda-s} & s < \lambda \\ \text{not defined} & \text{otherwise} \end{cases}$$



② $X \sim \text{Poisson}(\lambda)$

$$M_X(s) = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda} e^{sx}$$

$$= e^{-\lambda} \left[\sum_{x=0}^{\infty} \frac{(\lambda e^s)^x}{x!} \right] = e^{-\lambda e^s}$$

$$= e^{-\lambda(e^s - 1)}$$

③ $X \sim \text{Bernoulli}(p)$.

$$M_X(s) = e^{s \cdot 0} (1-p) + p e^{s \cdot 1}$$

$$= (1-p) + p e^s$$

④ $X \sim N(0, 1)$.

$$\begin{aligned} M_X(s) &= \int_{-\infty}^{\infty} e^{sx} e^{-\frac{x^2}{2}} \frac{1}{\sqrt{2\pi}} dx \\ &= e^{\frac{s^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{(x^2 - 2sx + s^2)}{2}} \frac{1}{\sqrt{2\pi}} dx \\ &= e^{\frac{s^2}{2}} \end{aligned}$$

Properties of MGF

① $Y = aX + b$

$$\begin{aligned} M_Y(s) &= E[e^{s(aX+b)}] \\ &= e^{sb} M_X(as) \end{aligned}$$

② $Y = X_1 + X_2$ X_1, X_2 independent

$$M_Y(s) = M_{X_1}(s) M_{X_2}(s)$$

$$E[e^{sY}] = E[e^{s(X_1 + X_2)}]$$

$$= E[\underbrace{e^{sX_1}} \cdot \underbrace{e^{sX_2}}]$$

$$= E[e^{sX_1}] E[e^{sX_2}] = M_{X_1}(s) M_{X_2}(s)$$

X_1, X_2 are independent

$$E[f(X_1)g(X_2)] = E[f(X_1)]E[g(X_2)]$$

$Y = X_1 + \dots + X_n$ are independent

$$M_Y(s) = \prod_{i=1}^n M_{X_i}(s)$$

$Y \sim \text{Binomial}(n, p)$

X_i 's are iid $\text{Bernoulli}(p)$

$$M_Y(s) = (M_{X_i}(s))^n \\ = ((1-p) + pe^s)^n$$